Development Strategy for a Competition RC Aircraft: Calculate, Simulate Selectively, Build Early, Test Often

1. Recommendation

I recommend a **hybrid approach**: do quick calculations and a little free-software simulation first, build a cheap prototype within about 10 days, fly it and measure it, then use the measurements to correct the simulation before building the final competition plane. **Why:** with four weeks and little money, the scarce resource is not computer time, it is the number of test flights we can make. Simulation cannot tell us whether the plane was built well, and building cannot try hundreds of ideas. Using each for what it does best beats using either alone.

2. Assumptions (replace with your real rules)

- **Mission:** carry a payload and fly circuits from a short runway.
- **Team and money:** 4 to 5 students, a 4-week deadline, enough money for two airframes plus spares.
- **Resources:** free software (XFLR5, OpenVSP), a student licence for CAD (computer-aided design, for example Fusion 360), a 3D printer or foam/balsa workshop, and a flying field on weekends.
- **Aircraft:** electric, sea-level air, airflow conditions typical of small RC planes (Reynolds number about 2×10^5 to 3×10^5).

3. The Two Approaches in Plain Language

Simulation-led means designing on the computer first, then building once. Tools: XFLR5 and OpenVSP (wing and stability), CAD such as Fusion 360 or SolidWorks (layout and weight), and ANSYS (detailed airflow, called CFD).

- **Good:** trying an idea costs minutes and no materials; it finds sizing and stability mistakes early; it explains *why* a design works; there is no crash risk.
- **Limits:** answers are only as good as the inputs (wrong weight in means wrong result out); models are simplified and do not know about a warped wing, loose linkages or a glue blob; advanced tools like ANSYS take skill and days; one big build means one chance.

Prototype-led means building a rough plane quickly, flying it, fixing it and repeating.

- **Good:** it includes all real physics, even things nobody thought of; it shows fit, wiring and vibration problems; the pilot gets practice; it is the only true proof that the plane flies.
- **Limits:** each build costs materials and days; weather decides when you can test; crashes break parts; results are noisy; without calculation you may not learn *why* something worked.

4. Comparison

These are judgments for a small student team, not measured data.

Criterion	Simulation-led	Prototype-led	**Hybrid (chosen)**

Criterion	Simulation-led	Prototype-led	**Hybrid (chosen)**
Cost	Free tools, but many engineer-hours; one expensive final build	Moderate per build, repeated; crashes cost parts	Free software plus two airframes, the first one cheap
Time	Days to weeks before anything flies	First flight in 1 to 2 weeks	Calculations in 5 days, first flight about day 12
Accuracy	Good for trends, weaker for exact numbers	Real, but noisy (wind, pilot)	Model corrected by measurements
Iteration speed	Minutes per change	Hours to days per change	Simulate many ideas, build the best few
Failure detection	Early sizing and stability errors; blind to build defects	Finds everything real, but only after building	Cheap errors on the computer, real ones on the ground and in the air
Final performance	Optimised on paper, untested	Robust, maybe not optimised	Robust and reasonably optimised

5. Why the Hybrid Wins

1. **Simulate what is cheap and well understood:** wing size, balance, motor choice.
2. **Test early what a computer cannot know:** real weight, thrust, build quality, strength, pilot feel.
3. **Use measurements to correct the model,** so its predictions for the second plane can be trusted.

Why not full CFD (ANSYS)? It takes expertise and hours per run, and it is hard to trust at this small scale. Our time is better spent flying. Use it only for a specific question that XFLR5 cannot answer.

6. Workflow: From Requirements to Refinement

1. **Read the rules.** Note the scoring, limits and constraints.
2. **Do hand calculations.** Weight budget, wing area, thrust and battery.
3. **Choose the concept and airfoil.** Use XFLR5/XFOIL polars.
4. **Simulate stability and performance.** Use XFLR5, OpenVSP and a motor calculator.
5. **Gate A (go/no-go).** Is the static margin 8 to 15%, and are stall speed, takeoff and endurance on target? If no, go back to step 2 or 3. If yes, continue.
6. **Lay out the design in CAD (Fusion 360).** Check CG and fit, and produce drawings and cut/print files.
7. **Build prototype P1.** Use cheap foam, balsa or printed parts.
8. **Run ground tests.** Check weight, CG, static thrust and control directions.
9. **Gate B (go/no-go).** Is the weight within 10% of budget and the thrust at or above target? If no, fix it before flying. If yes, continue.
10. **Run flight tests with test cards.** Progress from glide, to light payload, to full payload.
11. **Compare predicted vs. measured.** Calibrate the model.
12. **Refine the design and build the competition airframe C1.**

13. **Test C1, practise, freeze the design and pack spares.**

Why gates? A gate is a go/no-go checkpoint with numbers. It stops us spending days building a design that fails a five-minute calculation, or flying an aircraft that fails a two-minute thrust test.

Worked example: sizing from the rules (assumed numbers)

- **Target:** 4.0 kg takeoff mass (weight 39.2 N) and a stall speed (the slowest speed before the wing stops lifting) of 11 m/s or less. Assume the real wing reaches a maximum lift coefficient of 1.6.
- **Wing area:** $2 \times 39.2 / (1.225 \times 11^2 \times 1.6) \approx$ **$0.33\ m^2$**, about 12 kg per m^2 of wing.
- **Takeoff run:** lift off at about $1.2 \times 11 = 13.2$ m/s. With an average thrust of 14 N the acceleration is about 3.1 m/s^2, so the run is $13.2^2 / (2 \times 3.1) \approx$ **28 m**. Allow about 35 m for drag and soft ground.
- **Thrust needed:** propeller thrust drops as speed rises, so average thrust is roughly 70% of static thrust (a rule of thumb to be tested). That means about **20 N static thrust**, half the aircraft's weight.

Ten minutes of arithmetic fixes the wing size, the motor class, and the test the plane must pass before flying.

7. What to Simulate and What to Check on the Real Aircraft

Simulate (cheap and well understood):

1. **Airfoil lift and drag** at our airflow conditions (XFLR5/XFOIL). It picks the wing shape in minutes.
2. **Balance and stability** (XFLR5/OpenVSP): where the centre of gravity (CG, the balance point) must sit and how big the tail must be. Errors here cause crashes.
3. **Motor, propeller, battery and flight time** (motor calculator or spreadsheet). Choose parts before buying them.

Check on the real aircraft:

1. **Actual weight and CG.** Glue, wires and paint are never in the model, and weight sets the stall speed.
2. **Static thrust** (pull force with the plane held still) **and battery current.** Real parts differ from calculators by 10 to 20%.
3. **Takeoff distance and stall speed.** They depend on everything at once.
4. **Wing strength.** Load-test with sandbags (about 2× the flight load, assumed). A computer cannot see a weak glue joint.

8. Checking the Simulation Against Reality

Validation means asking whether the simulation matches the real world. Compare predicted and measured values:

Quantity	How to measure	Accept if within
Weight	Kitchen or hanging scale	±5%

Quantity	How to measure	Accept if within
CG	Balance on two fingers	±5 mm
Static thrust	Scale plus current meter	±15%
Stall speed	GPS logger, flown with and against the wind and averaged	±10%

Procedure: measure, compare, and if the result is outside the range find which input was wrong (usually weight, drag or thrust). Fix the model, note the change, and re-run. The final test: if the corrected model correctly predicts how the second plane differs from the first, we can trust it.

9. The Prototype Won't Take Off

A takeoff needs three things: enough **thrust** to speed up, enough **speed** to make lift, and enough **lift and nose attitude** to rotate. The diagnosis finds which one is missing.

Likely causes: overweight; weak thrust (wrong propeller, sagging battery voltage, propeller on backwards); too little lift (wing angle too low, warped wing, real lift lower than assumed); wrong balance or control (CG too far forward, elevator too small or reversed); extra drag (soft grass, big wheels, tailwind); or the simulation was given wrong inputs.

Diagnostic plan, cheapest first:

1. **Make it safe and run a five-minute checklist:** controls move the right way, CG is correct, nothing rubs, battery is charged, propeller is on correctly.
2. **Weigh it** and compare with the budget. More than 10% over is a prime suspect.
3. **Static thrust test** against the 20 N target. If it is low, fix propulsion first.
4. **Filmed taxi test** against marked distances. Does it reach about 1.2 × stall speed before the runway ends? If it never reaches speed, the problem is thrust, drag or ground. If it reaches speed but won't lift, check wing angle, elevator and CG. If it lifts and then sinks, check balance, trim and stall.
5. **Hand-launch glide test.** If it glides well, the wing and balance are fine and the problem is thrust or ground.
6. **Change one thing at a time.** Then feed the measured weight, thrust and lift back into the simulation so it explains what happened.

10. Four-Week Schedule With Contingency

Week	Plan	Checkpoint
1	Rules, calculations, concept, simulation, CAD. **Order motor, motor controller, battery and servos on day 2.** Start P1 parts.	Gate A by day 5. Day 7 is a buffer.
2	Build P1, ground tests, first flights (glide, then light, then payload).	Gate B before flying. Day 14 is a weather buffer.
3	Analyse flights, correct the model, refine, build competition plane C1.	C1 ready by day 19. Day 21 is a buffer.

Week	Plan	Checkpoint
4	Test C1 with full payload, practise the mission, **freeze the design by day 25**, pack spares.	Days 27 and 28 reserved, no changes.

Contingency is about 25% of the time: one buffer day per week plus the last two days. If we fall behind, **cut features, not quality.** Main risks and answers: late parts (order early), bad weather (do ground tests), crashes (cheap P1, spare parts), battery faults (spare battery), a missing teammate (named backups).